

Looking into space



Eye in the sky

The planets, stars, and galaxies have been revealed to us by telescopes and space probes. The Hubble Space Telescope has been a constant eye on the universe since 1990.

To the edge of the visible universe

The universe is here separated into five steps, moving farther away from our home planet. From Earth, we move to the Moon, then the solar system, the Milky Way, and the Local Group of galaxies. Beyond this, however deep we look into space, we find more galaxies.

WINDOW ON THE UNIVERSE

The sky of our home planet, Earth, is our window on the universe. As we look away out into space, we see some of its objects. In the daytime, there is just one large object, the Sun. It is our local star, and its brilliance fills the sky.

On a cloud-free night, the dark sky is full of starry pinpoints of light. With the naked eye we can see more than 2,000 of these objects, and with binoculars the number rises to over 40,000. Looking carefully, we can discern that some are not

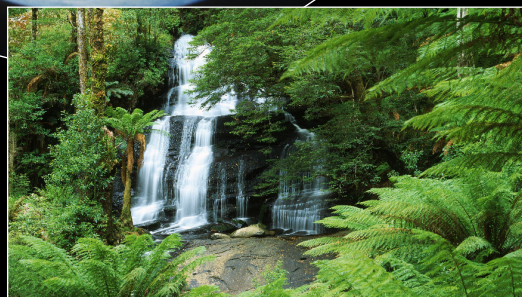
bright points of light but disk-shaped planets, while the fuzzy patches we see can be galaxies full of stars.

The objects in the universe are so remote that it is difficult to compare their sizes and distances. They all appear to be an equal distance from Earth. In reality, though, they are not only at varying distances from us, they are also vastly different distances from each other.

Our "neighborhood" includes the Sun, Moon, and planets. Beyond is the realm of the stars, the Milky Way Galaxy. And outside of this are more galaxies overflowing with stars.

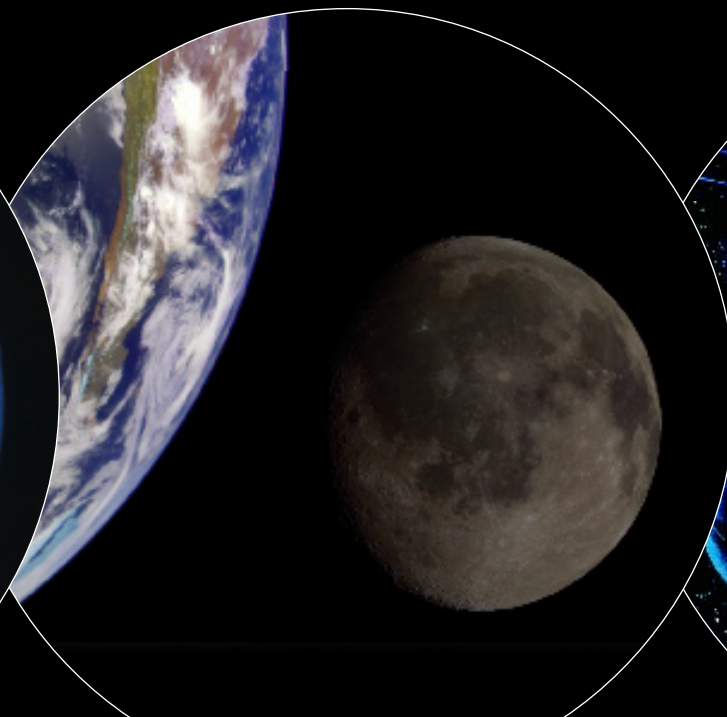
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FINDING YOUR WAY



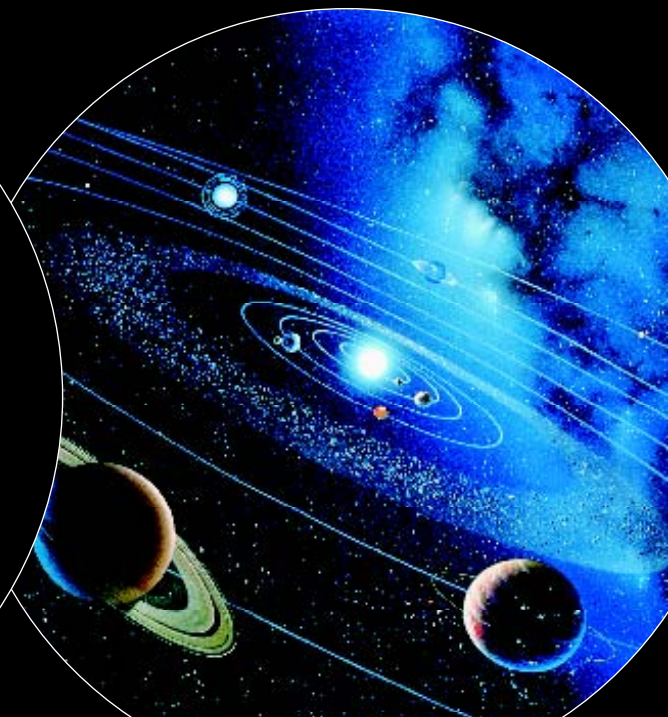
Earth: our home planet

A blue ball of rock 7,926 miles (12,756 km) across, Earth is the only place in the universe known to have life; and uniquely, much of it is covered by liquid water. As Earth spins and moves in space, we are able to look out to different parts of the universe.



The Moon

Closest to Earth, and a familiar feature of its sky, is the Moon. At its brightest, it is the next brightest object in our sky after the Sun. This ball of rock accompanies Earth in space, traveling around us every 27.3 days. It is a dry, dead world covered in impact craters formed billions of years ago. The Moon is the only place that man has walked apart from Earth.



The solar system

Earth and the Moon are part of the solar system (above, top), which is a volume of space that includes billions of objects. The central, most massive object is the Sun, and all other members of the system travel around it. Next biggest are the eight planets, and the giant of these is Jupiter (above). Comets and asteroids are more numerous but smaller.